

When Users Invoke AI in Public Discussions: Interactional Patterns and Depth of AI-Mediated Knowledge Construction on Social Media

Yi Tao*

*School of Educational
Information Technology
South China Normal University
Guangzhou, China
taoyi@m.scnu.edu.cn
Corresponding author*

Zimeng Wang

*School of Public Administration
South China Normal University
Guangzhou, China
wzimeng0304@outlook.com*

Jiahao Miao

*School of Educational
Information Technology
South China Normal University
Guangzhou, China
miaojiahao@m.scnu.edu.cn*

Boxun Yang

*School of Educational
Information Technology
South China Normal University
Guangzhou, China
2292797881@qq.com*

Ruilin Lai

*School of Educational
Information Technology
South China Normal University
Guangzhou, China
ruilin@m.scnu.edu.cn*

Jinshuo Dong

*School of Educational
Information Technology
South China Normal University
Guangzhou, China
2024020869@m.scnu.edu.cn*

Abstract—As conversational AI accounts become embedded in public social media, users increasingly invoke them not only to obtain answers but also to interpret content, evaluate evidence, and intervene in ongoing disputes. While recent studies have described the roles and uses of AI accounts in public discourse, less is known about whether and how such interactions form observable processes of knowledge construction. Drawing on computer-supported collaborative learning research and Gunawardena et al.'s Interaction Analysis Model, this study analyzed 869 knowledge-related AI-invocation episodes from rehydrated GrokSet conversations on X. We identified six interactional patterns through which AI entered public discussion, coded the highest observable level of knowledge construction in each episode, and examined how knowledge construction depth varied across patterns. Results show that public AI-mediated knowledge work was dominated by terminal dyadic consultation, in which a user asks AI for information, explanation, or judgment and the visible thread largely ends after the AI response. Accordingly, most episodes remained at KC1, sharing and comparing information (80.55%). However, deeper knowledge construction emerged under specific interactional conditions. Verification and argumentative intervention episodes were strongly associated with KC4, involving evidence-based testing and modification of claims, while post-response uptake and branching episodes were the main site of KC3, involving negotiation and co-construction of meaning. These findings suggest that public AI invocations do not automatically produce collaborative knowledge construction. Rather, their epistemic value depends on how AI responses are positioned, taken up, and contested within human-AI-human interaction.

Keywords—*AI-mediated knowledge construction; social media; human-AI interaction; AI invocation; public discourse; Interaction Analysis Model*

I. INTRODUCTION

Collaborative knowledge construction has long been an important issue in online learning research [1]. Today, public

social media provides a new setting for this process, as users increasingly bring conversational AI accounts into open discussions. Here, AI invocation refers to the increasingly common practice of bringing an AI account into a public thread by tagging or mentioning it, such as @grok on X or @doubao in Douyin comment sections. These practices can be approached as an informal form of human-AI collaborative knowledge construction, in which AI responses are embedded in open, multi-party, and socially visible discussions. Unlike traditional online learning communities or private dyadic human-AI learning, AI responses may become shared resources that other users quote, challenge, ignore, or use to continue the discussion.

Recent studies have begun to describe this emerging form of public human-AI interaction. Bobek et al. [2] examined Grok as a form of AI-assisted sensemaking on X, showing that users often invoke it to obtain information, clarify content, or verify claims within ongoing conversations. Mei et al. [3] further characterized Grok's social roles and uses, finding that it is frequently used for information seeking, fact-checking, argument support, and debate, while also being positioned as an information provider, truth arbiter, advocate, or adversary. Using GrokSet, Migliarini et al. [4] extended this line of research by analyzing large-scale, public, multi-party human-LLM threads, thereby moving beyond isolated prompt-reply pairs to examine broader interactional contexts. Although these studies reveal how Grok is used and what roles it may play in public discourse, they do not directly examine public AI invocation as a form of knowledge construction.

To address this gap, this study examines 869 knowledge-related AI-invocation episodes in public social media discussions. It analyzes how AI is invoked and taken up in public interaction, what levels of knowledge construction are visible in the discourse, and how knowledge construction depth varies across interactional patterns. By shifting attention from isolated user-AI exchanges to public human-AI-human

interaction, this study seeks to clarify when AI functions merely as an answer provider and when it becomes part of evidence testing, meaning negotiation, and socially visible knowledge construction.

II. RELATED WORK

A. Online Collaborative Knowledge Construction

Research on online collaborative knowledge construction emphasizes that learning in networked environments develops through discourse: participants make ideas visible, compare perspectives, negotiate meanings, and test emerging understandings [5]. This view is consistent with CSCL research, which treats knowledge as something constructed through socially organized interaction, intersubjective meaning making, knowledge-building communities, and argumentative knowledge construction [6].

These frameworks are useful for the present study because AI-invocation episodes also leave visible discourse traces: users invoke AI, AI responds, and other participants may ignore, quote, challenge, or build on the response. Gunawardena et al.'s Interaction Analysis Model (IAM) is especially relevant because it distinguishes multiple levels of knowledge construction, from information sharing and dissonance to negotiation, evidence testing, and application. It therefore provides an analytic lens for examining what level of knowledge construction is observable in public human-AI-human interaction.

B. Generative AI and Knowledge Construction

Generative AI introduces new questions for knowledge construction research because large language models can participate directly in discourse by explaining concepts, summarizing information, generating arguments, and evaluating claims. From a CSCL perspective, this raises the question of whether AI should be understood as collaborator, tool, or conversational resource in knowledge-building processes [7].

Empirical studies show that human-AI collaboration varies by interactional arrangement: AI can reorganize group problem solving [8], support argumentative co-writing [9], or be passively reused without much negotiation. Thus, the educational significance of AI depends not only on response quality, but also on how participants position and take up AI contributions. However, much of this work has focused on designed learning settings or task-based collaboration, leaving open the question of how AI-mediated knowledge construction unfolds in open, public, and socially visible discussions.

C. Public AI Invocation on Social Media

Public AI invocation connects these two lines of research. Compared with private chatbot use, AI invocation on social media is public, threaded, and embedded in ongoing discussion. Users do not simply interact with AI in isolation; rather, they bring AI responses into a visible interactional space where other participants may evaluate, challenge, ignore, or reuse them. In this sense, AI responses may answer the invoking user, but they can also become shared objects for public sensemaking, evidence evaluation, and further discussion.

Existing work has not yet examined whether public AI-invocation episodes function as observable knowledge construction processes. It remains unclear what interactional patterns emerge, what knowledge construction levels can be observed, and how depth varies across patterns. Therefore, this study asks:

RQ1: What interactional patterns of AI-mediated knowledge construction emerge in knowledge-related AI-invocation episodes?

RQ2: What levels of knowledge construction are observable in knowledge-related AI-invocation episodes?

RQ3: How do knowledge construction levels vary across interactional patterns?

III. METHODS

Across data screening and coding, this study adopted an LLM-assisted procedure with double human verification. The LLM was used only to generate preliminary screening and coding suggestions. Two researchers then independently reviewed the preliminary labels for episode screening, interactional pattern analysis, and IAM-based knowledge construction depth coding. To assess coding reliability, Cohen's kappa was calculated before consensus discussion, and the kappa values exceeded 0.80 across the main coding tasks. Cases of disagreement or ambiguity were discussed and resolved through consensus. Therefore, the final analytic labels reported in this study were based on human verification rather than LLM outputs alone.

A. Data Source and Screening

This study uses GrokSet, a large-scale public dataset of X conversation threads involving the official @grok account, downloaded from its Hugging Face repository [10]. The dataset spans March to October 2025 and contains 182,707 conversation threads and 1,098,394 tweets. Because GrokSet is released in a dehydrated format, containing tweet IDs, annotations, and structural metadata rather than fully available tweet texts, the textual content was rehydrated through TwitterAPI.io [11]. Only tweets that were still publicly accessible at the time of rehydration were retained. The rehydrated data were then merged at the episode level, preserving user prompts, AI responses, and relevant surrounding replies for subsequent analysis.

The unit of analysis is an AI-invocation episode, defined as a thread segment containing a user's AI invocation, the AI response, and relevant surrounding replies needed to interpret the interaction. Following Gunawardena et al.'s view of online knowledge construction as meaning negotiation and knowledge co-construction, this study retained only knowledge-related episodes. These were episodes focused on identifiable information, content, evidence, explanation, viewpoint, or claim, and involving a knowledge-oriented purpose such as understanding, explaining, verifying, judging, evaluating, supporting, or challenging that object. Episodes mainly oriented toward entertainment, image generation, casual chat, emotional expression, harassment, adult-content identification, or jailbreak testing were excluded. From 1,285 complete AI-invocation episodes containing both a user prompt and an AI

response, 869 knowledge-related episodes were finally retained for coding and analysis.

B. Interactional Pattern Analysis

To answer RQ1, this study conducted a theory-informed inductive analysis of the interactional patterns in the included AI-invocation episodes. Here, interactional patterns refer to episode-level configurations of AI-mediated public interaction, including how AI was invoked, how its response was taken up, and how subsequent human-AI-human discourse unfolded. The analysis was informed by interaction analysis and content analysis in computer-mediated discussion, which treat visible participation, reply relations, and discourse sequences as traces of online interaction [6].

Based on this perspective, the analysis examined how an AI response was embedded in the public thread, including the pre-response context, post-response uptake, human-human relations, human-AI relations, and the complexity of the interactional sequence. Recurrent interactional patterns were first identified from visible discourse moves and thread relations. Similar patterns were then compared, merged, and refined into dominant pattern categories. The final analysis reports the frequency and proportion of each pattern and uses these categories to examine how observable knowledge construction depth varies across interactional patterns.

C. Knowledge Construction Depth Analysis

To answer RQ2, each retained episode was coded for the highest observable level of knowledge construction based on Gunawardena et al.'s Interaction Analysis Model (IAM) [5]. IAM was developed to examine the social construction of knowledge in computer-mediated discussion and conceptualizes knowledge construction as a five-phase process: sharing and comparing information, discovering and exploring dissonance, negotiating meaning and co-constructing knowledge, testing and modifying proposed constructions, and reaching agreement or applying newly constructed meaning. In this study, these IAM categories were adapted to public AI-invocation episodes to capture visible discourse moves rather than unobservable learning outcomes.

Single-turn question-answer episodes were coded as KC1 when they involved information sharing or explanation, rather than being treated as no knowledge construction. Higher levels required observable dissonance, negotiation, evidence-based testing, modification, agreement, or application in the thread. If an episode showed multiple levels of knowledge construction, the highest observable level was recorded.

D. Cross-Pattern Association Analysis

To answer RQ3, the finalized knowledge construction depth codes were cross-tabulated with the interactional pattern categories identified in RQ1. Row percentages were used to describe how knowledge construction levels were distributed within each interactional pattern. Chi-square tests, permutation tests, and Cramer's V were used as descriptive association indicators, not as evidence of causal relationships.

TABLE I. KNOWLEDGE CONSTRUCTION DEPTH CODING BASED ON IAM

Code	IAM category	Operational criterion
KC1	Sharing and comparing information	Users or AI provide facts, descriptions, explanations, summaries, translations, or comparisons.
KC2	Discovering and exploring dissonance	The episode shows disagreement, conflicting claims, error identification, uncertainty, or competing interpretations.
KC3	Negotiating meaning and co-constructing knowledge	Participants clarify, compare, elaborate, reconcile, or negotiate interpretations.
KC4	Testing and modifying proposed constructions	Participants use evidence, sources, examples, counterexamples, or reasoning to test, modify, or refine claims.
KC5	Reaching agreement or applying newly constructed meaning	Participants summarize, accept, apply, or extend a newly constructed understanding to a case or conclusion.

IV. RESULTS

This section reports the results for the three research questions. The analysis is based on 869 knowledge-related AI-invocation episodes that contained both a user invocation and an AI response. The results are organized in three parts. First, we describe the interactional patterns through which AI entered public discussion. Second, we report the highest observable level of knowledge construction in these episodes using the adapted IAM coding scheme. Third, we examine how knowledge construction depth varies across interactional patterns.

A. RQ1: Interactional Patterns of AI-Mediated Knowledge Construction

The inductive analysis identified six substantive interactional patterns, plus a small uncertain category. Fig. 1 summarizes the distribution of these patterns. The most common pattern was terminal dyadic knowledge consultation (IP1), which accounted for 602 episodes, or 69.28% of the sample. These episodes typically consisted of a user's direct knowledge-oriented prompt followed by an AI response, with no visible continuation in the public thread. In these cases, AI was mainly positioned as a direct source of information, explanation, summary, or advice.

A second set of patterns involved AI being attached to an already visible public object, issue, or claim. Referenced-content sensemaking (IP2) accounted for 63 episodes (7.25%). In these episodes, users asked AI to interpret a post, video, image, chart, link, statement, or thread. AI was therefore not simply answering an abstract question; it was used to make sense of a shared object in the public discussion. Verification and evidentiary adjudication (IP3) accounted for 68 episodes (7.83%). These episodes positioned AI as a fact-checker or truth evaluator, asking it to judge whether a claim, image, statistic, report, or public statement was true, accurate, fake, misleading, or supported by evidence.

A smaller group of episodes placed AI directly into human disagreement or practical problem solving. Argumentative third-party intervention (IP4) appeared in 16 episodes (1.84%). In these cases, AI was invoked to compare competing claims, support or refute an argument, evaluate another participant's position, or provide evidence within an ongoing disagreement.

Procedural problem-solving consultation (IP5) appeared in 46 episodes (5.29%). These episodes focused on rules, eligibility, deadlines, recruitment, exams, school policies, fees, loans, or other action-oriented questions. AI was used as a practical knowledge resource rather than only as a general explainer.

Finally, post-response uptake and branching (IP6) appeared in 54 episodes (6.21%). These cases were especially important because the AI response did not simply terminate the episode. Instead, visible subsequent interaction continued after the AI response, through follow-up questions, additional comments, redirection, or public continuation of the topic. Twenty episodes (2.30%) were marked uncertain because the visible prompt or context was too thin or noisy to assign a stable interactional pattern. Overall, RQ1 shows that knowledge-related AI-invocation episodes are dominated by short user-AI exchanges, but a smaller subset embeds AI in richer human-AI-human configurations where AI becomes an interpretive object, evidentiary resource, argumentative participant, or pivot for further discussion.

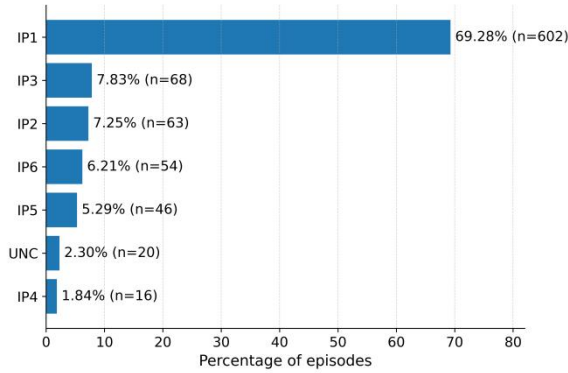


Fig. 1. Distribution of interactional patterns.

B. RQ2: Observable Knowledge Construction Depth

RQ2 examined the highest observable level of knowledge construction in each retained episode. KC1, sharing and comparing information, was the dominant level of observable knowledge construction, appearing in 700 episodes (80.55%). These episodes usually involved AI providing facts, explanations, summaries, translations, comparisons, or practical information without visible negotiation or testing. This finding indicates that most public AI-invocation episodes functioned primarily as information-sharing exchanges.

KC2, discovering and exploring dissonance, was observed in 13 episodes (1.50%). These episodes surfaced uncertainty, contradiction, disagreement, or competing claims, but did not show enough visible evidence testing or interactive negotiation to be coded at a higher level.

KC3 episodes typically involved visible follow-up, clarification, comparison, elaboration, or continuation around the same knowledge object. Although infrequent, these episodes were important because the AI response became part of a continuing interpretive process rather than a terminal answer.

KC4, testing and modifying proposed constructions, was observed in 109 episodes (12.54%). This was the second largest category after KC1. KC4 episodes usually involved

fact-checking, evidence evaluation, source-based reasoning, statistics, official documents, examples, or other visible forms of testing a claim or interpretation. In these cases, AI was not only providing information but was also helping evaluate whether a proposed claim or explanation could be supported.

KC5, reaching agreement or applying newly constructed meaning, was rare, appearing in only one episode (0.12%). This indicates that explicit public acceptance, agreement, or application of a newly constructed understanding was difficult to observe in short social media AI-invocation episodes. Twenty episodes (2.30%) remained uncertain because the available text did not support stable IAM-level coding.

Overall, RQ2 shows that most episodes remained at the level of information sharing, while evidence-based testing represented the most common form of deeper knowledge construction. Negotiation and co-construction were less frequent and usually required visible continuation after the AI response.

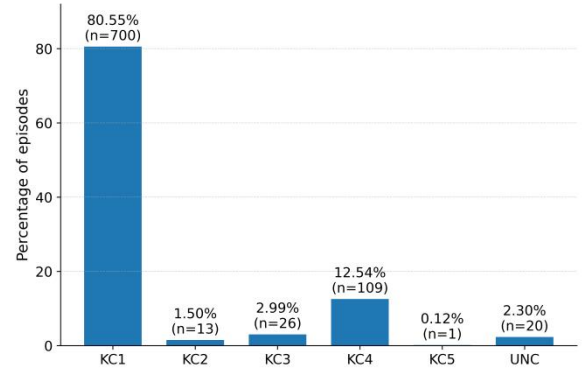


Fig. 2. Distribution of knowledge construction depth.

C. RQ3: Knowledge Construction Depth Across Interactional Patterns

To answer RQ3, the RQ1 pattern codes were cross-tabulated with the RQ2 IAM-based knowledge construction codes. The main association analysis excluded the 20 uncertain cases, leaving 849 episodes. The 6-by-5 cross-tabulation showed a clear descriptive association between interactional pattern and observable knowledge construction depth, Pearson chi-square = 942.533, df = 20, Cramer's V = 0.527. Because the table was sparse, a permutation test with 20,000 shuffles was also used, yielding $p < 0.001$. A collapsed robustness check comparing KC1 with KC2-KC5 also showed a strong association, chi-square = 566.778, df = 5, Cramer's V = 0.817, permutation $p < 0.001$. These statistics should be interpreted as descriptive association indicators rather than evidence of causal relationships, because both variables were derived from visible interactional evidence.

The row percentages in Fig. 3 show where the association came from. IP1 terminal dyadic consultation was overwhelmingly concentrated at KC1: 96.51% of IP1 episodes were coded as KC1. IP2 referenced-content sensemaking and IP5 procedural problem-solving consultation were also mainly KC1, at 92.06% and 91.30%, respectively. These patterns generally involved AI supplying an explanation, interpretation, or practical answer without visible subsequent negotiation.

By contrast, IP3 verification and evidentiary adjudication was strongly associated with KC4. Among IP3 episodes, 91.18% were coded as KC4. Similarly, within the small category of IP4 argumentative third-party intervention, 81.25% of episodes were coded as KC4. This suggests that when AI was invoked as a fact-checker, evidence evaluator, or argumentative resource, the observable knowledge work was more likely to involve testing claims with evidence, sources, data, or reasoning. IP6 post-response uptake and branching had a different profile: only 35.19% of IP6 episodes were KC1, while 46.30% reached KC3. This makes IP6 the main site of visible meaning negotiation and co-construction, because the AI response became part of a continuing public interaction rather than simply ending the exchange.

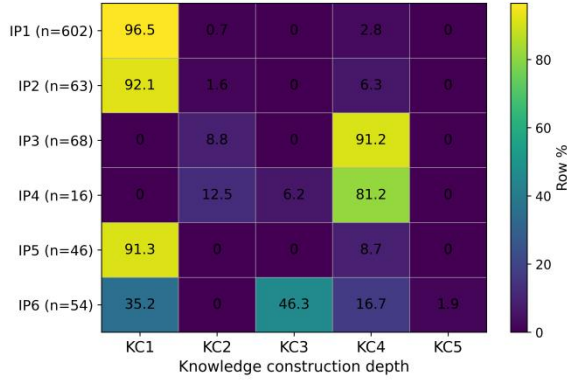


Fig. 3. Knowledge construction depth by interactional pattern.

Taken together, RQ3 shows that knowledge-related AI-invocation episodes did not all support the same kind of knowledge construction. Most episodes represented public information sharing between a user and AI. Deeper knowledge construction became visible under more specific interactional conditions: evidence testing was concentrated in verification and argumentative patterns, while meaning negotiation was concentrated in episodes where the AI response was followed by visible public uptake. Thus, AI-mediated knowledge construction on social media appears to be shaped not only by what users ask AI, but also by how the AI response is embedded in the surrounding public interaction.

V. DISCUSSION

This study examined public AI-invocation episodes as an emerging form of AI-mediated knowledge construction on social media. Rather than treating user-AI exchanges as isolated prompts and responses, the analysis focused on how AI responses were embedded in public interactional patterns and what levels of knowledge construction were observable from the visible discourse. Three main findings stand out.

A. Public AI Invocation Is Mostly Consultative, but Still Knowledge-Oriented

The first finding is that knowledge-related AI-invocation episodes are dominated by terminal dyadic knowledge consultation. In nearly seven out of ten episodes, a user invoked AI to obtain information, explanation, interpretation, or advice, and the visible thread did not continue after the AI response. This pattern suggests that many public AI invocations resemble open, socially visible question-answer exchanges. They are public rather than private, but

interactionally they often remain close to a single-turn consultation.

This finding should not be interpreted as an absence of knowledge construction. Following the adapted IAM framework, single-turn information sharing and explanation were coded as KC1 because they introduce knowledge resources into the public discussion. However, the dominance of KC1 also indicates that most episodes did not show visible negotiation, revision, or application of knowledge. In other words, public AI invocation often makes information available, but it does not automatically produce deeper collaborative knowledge construction.

B. Interactional Patterns Shape Observable Knowledge Construction Depth

The second finding is that knowledge construction depth varied strongly across interactional patterns. Terminal dyadic consultation, referenced-content sensemaking, and procedural problem-solving were mostly associated with KC1. These patterns often involved AI supplying an answer, summary, interpretation, or practical guidance. By contrast, verification and argumentative intervention were strongly associated with KC4. When users invoked AI to check whether a claim was true, evaluate evidence, or respond to a disagreement, the visible knowledge work more often involved testing proposed constructions through sources, data, examples, or reasoning.

This pattern suggests that deeper knowledge construction is not simply a property of longer or more detailed AI responses. Instead, it depends on the interactional work that the AI response is asked to perform. When AI is positioned as a fact-checker, evidentiary adjudicator, or third-party argumentative resource, the episode is more likely to move beyond information sharing toward evidence-based testing. This extends prior work on public AI use by showing not only that users ask AI for information or verification, but also that different ways of positioning AI correspond to different observable levels of knowledge construction.

C. Post-Response Uptake Enables Visible Co-Construction

The third finding is that KC3, negotiation and co-construction of knowledge, was concentrated in post-response uptake and branching episodes. These cases were relatively uncommon, but they are theoretically important because they show when AI-mediated knowledge construction becomes visibly social. In such episodes, the AI response did not simply terminate the interaction; instead, users continued the discussion, asked follow-up questions, redirected the topic, clarified prior claims, or built on the AI response.

This finding is important for educational technology because it suggests that the social value of AI in discussion spaces may depend less on whether AI provides a correct answer alone and more on whether the AI response becomes a shared object for further interaction. A high-quality AI response may remain interactionally terminal if no one takes it up. Conversely, even a short AI response can become part of a richer knowledge-building sequence if participants use it as a basis for clarification, comparison, challenge, or extension.

D. Implications for Educational Technology Research and Design

The findings have three implications for educational technology research. First, public AI invocation can be understood as an informal extension of computer-mediated knowledge construction in open social media settings. Although these episodes should not be treated as direct evidence of formal learning, they reveal how people mobilize AI in everyday knowledge work.

Second, AI should not be conceptualized only as a tutor or answer generator. In public discussion, it may function as an interpreter, fact-checker, evidentiary resource, argumentative third party, or catalyst for continued discussion. This means that the educational significance of AI depends not only on response quality, but also on how AI responses are positioned within discourse sequences.

Third, AI-supported discussion design should attend to uptake. If educators or platform designers want deeper knowledge construction, they need to create conditions in which AI responses are compared with evidence, questioned by peers, connected back to prior claims, or used as a basis for further reasoning. Such activities can shift public AI use from answer retrieval toward accountable sensemaking, evidence evaluation, and collaborative interpretation.

The findings also suggest a methodological implication. In public social media, the analytically meaningful unit is not only the user prompt or the AI reply, but the whole episode in which a prompt, response, context, and possible uptake are connected. Treating the episode as the unit of analysis makes it possible to distinguish an answer that terminates interaction from an answer that becomes a resource for further questioning, dispute, or sensemaking.

E. Limitations and Future Research

This study has several limitations. First, the analysis relies on rehydrated public X data, so deleted, private, unavailable, or non-returned tweets could not be examined. It also focuses on GrokSet, one platform, and one AI assistant account; therefore, the findings should be interpreted as a case of the broader phenomenon of public AI invocation rather than as evidence generalizable to all platforms, languages, or AI systems. Second, the coding captures visible discourse rather than individual learning outcomes, belief change, or long-term knowledge development. Although LLM-assisted coding was double-verified by human researchers, ambiguity may remain in multilingual, short, ironic, or context-dependent posts.

Finally, this study is exploratory and mainly uses descriptive statistics, cross-pattern comparison, and association tests to identify basic patterns of AI-mediated knowledge construction. The observed association between interactional pattern and knowledge construction depth should therefore be interpreted descriptively rather than causally. Future work could combine larger and more diverse samples with social network analysis, sequence analysis, and closer qualitative validation to examine how AI responses reshape reply structures, participant roles, interaction chains, and post-response uptake across platforms.

VI. CONCLUSION

This study analyzed 869 knowledge-related public AI invocation episodes to examine how AI-mediated knowledge construction unfolds in social media discussions. The findings show that most episodes are terminal dyadic consultations and remain at the level of information sharing. However, deeper forms of knowledge construction appear under specific interactional conditions. Verification and argumentative patterns are closely connected to evidence-based testing, while post-response uptake and branching provide the clearest setting for visible negotiation and co-construction.

Overall, the study suggests that public AI invocation is not a single uniform practice. Its epistemic value depends not only on what AI says, but also on how users position AI within public discussion and whether other participants take up, question, or extend the AI response. By shifting attention from isolated user-AI exchanges to human-AI-human interaction, this paper provides a conceptual bridge between social media sensemaking and AI-mediated knowledge construction.

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